

COSH: Closed-Form Onset Search via Hyperbolic Response

Methodology for pre-flight critical-moment identification of a multirotor from ground-contact tip-over excitation

This document specifies the complete identification methodology, from raw signals to the take-off feed-forward moment. Every stage mirrors the implementation in `critical_value_getter_pieewise.py`; the companion analyses (model-fidelity bounds, excitation design, ground-effect bracket) are documented in `fidelity_section.tex` and are only referenced here.

1 Problem setting and notation

A multirotor rests on its landing gear. Ramping the commanded roll (pitch) moment tilts the vehicle until it begins to rotate about the landing-gear contact line — the *pivot*. The applied moment at that *rotation onset* is the critical moment M_{crit} ; comparing the two tip directions yields the CoM offset and the feed-forward moment M_{ff} that compensates take-off. No rig, no motion capture and no stable in-flight data are used.

Symbol	Meaning	Value	Provenance
W, z_{CoM}	weight, CoM height	31.59 N, ~ 0.3 m	measured / conservative
$l_{p,\phi}, l_{p,\theta}$	pivot offsets (roll, pitch)	0.140, 0.110 m	conservative geometry
J_P	inertia about the pivot line	—	never used numerically
C_T	thrust map, $T_i = C_T \Omega_i^2$	1.3175×10^{-7}	bench calibration
M_{thr}	window threshold	0.01 N m	noise floor
M_{cap}	allocator caps (x, y)	2.37, 2.74 N m	allocation feasibility
ϵ_{max}	ramp-slope gate	3%	sensitivity chain (Sec. 5)
N_{min}	window-sample gate	38	30 pre + 8 post
$e_{\text{lin,max}}$	linearity-RMSE gate	30 mN m	$10\times$ noise floor
N_{seg}	sweep margin (each end)	8	degenerate-candidate guard

Table 1: Design constants. (C_2, K) are *calibrated*, not set (Sec. 7).

2 Pipeline overview

The complete pipeline is five stages; the three algorithms nest as Algorithm 3 $\supset$ Algorithm 2 $\supset$ Algorithm 1, and Algorithm 1 is the only stage that touches the angular rate.

1. **Signals and window** (Sec. 4) — per run: reconstruct $M(t)$ from the measured rotor speeds and take $\omega(t)$ from odometry; open the excitation window at $|M| > M_{\text{thr}}$ (backward from the peak), close it at the moment peak, truncated at the allocator cap.
2. **Run gates** (Sec. 5) — discard runs failing the slope-error (3%), sample-count (38) or linearity-RMSE (30 mN m) gates; moment trace only, so no circularity.
3. **Calibration, once per configuration and axis** (Sec. 7) — Algorithm 3 scans (C_2, K) coarse-to-fine; each candidate is scored by Algorithm 2, which runs Algorithm 1 on every gated run and sums the two directional CVs of M_{crit} across ramp rates.
4. **Identification, per run** (Sec. 6) — Algorithm 1 with the calibrated (C_2^*, K^*) pinned, at full sweep resolution: amplitude fixed by $C_1 = KM$, baseline by the pre-segment median, the onset index swept exhaustively; $M_{\text{crit}} = M(t_{j^*})$.

5. **Aggregation** (Sec. 8) — directional means $\bar{M}_\pm$, the pivot-free feed-forward moment $M_{\text{ff}} = \frac{1}{2}(\bar{M}_+ + \bar{M}_-)$, and the CoM offset for validation against the load-cell truth (Sec. 9).

3 Contact-phase dynamics and the closed form

About the active pivot line, with the small-angle attitude dependence retained ($\cos \varphi \approx 1$, $\sin \varphi \approx \varphi$), the roll/pitch dynamics read

$$J_P \dot{\omega} = M(t) + f l_p - W(l_p + \lambda_{\text{off}}) + W z_{\text{CoM}} \varphi, \quad (1)$$

where M is the commanded body moment, f the collective thrust and λ_{off} the CoM offset along the excited axis. Below the critical value the vehicle is static. At the onset the net moment vanishes, which supplies *two* anchoring conditions,

$$\omega(t_{\text{crit}}) = 0, \quad \dot{\omega}(t_{\text{crit}}) = 0. \quad (2)$$

Past the onset gravity feeds back positively: the linearized dynamics have *real* eigenvalues $\pm C_2$ and, with the measured moment ramp $M(t) = M_{\text{crit}} + \dot{M} \tau$ in onset-relative time $\tau = t - t_{\text{crit}}$, the exact solution of (1)–(2) collapses to the hyperbolic closed form

$$\omega(\tau) = C_1 (\cosh(C_2 \tau) - 1), \quad C_1 = K \dot{M}, \quad K = \frac{1}{W z_{\text{CoM}}}, \quad C_2 = \sqrt{\frac{W z_{\text{CoM}}}{J_P}} \quad (3)$$

for $\tau \geq 0$ (and $\omega = 0$ for $\tau < 0$). Because $J_P C_2^2 = W z_{\text{CoM}}$, the amplitude C_1 is *fixed* by the measured ramp rate: with (C_2, K) pinned as per-configuration effective constants and the baseline fixed by continuity, **no shape parameter is free per run**. The time-quadratic model of the original submission is the $C_2 \rightarrow 0$ limit of (3) — it discards the gravity term that creates the instability, with 14–75% truncation error over the windows actually used — and is retained only as a baseline.

4 Signals and excitation window

Per run, the moment is reconstructed from the measured rotor speeds, $M(t) = C_T \sum_i b_{a,i} \Omega_i^2(t)$, and $\omega(t)$ is taken from the odometry rate. The window end is $i_{\text{end}} = \arg \max_t |M|$, *truncated at the first sample exceeding the allocator cap* M_{cap} : past it the thrust allocation saturates, the executed ramp is no longer linear, and the cosh family does not apply. (On the reference dataset the cap never binds; peak $|M| \leq 1.7$ N m.) The window start is the *last* sub-threshold sample before the peak, walking backwards — robust to spin-up blips that cross the threshold long before the actual ramp.

5 Onset-free run gates

Before any onset is estimated, each run must pass three gates evaluated on the measured moment trace alone (hence free of circularity):

1. **Ramp-slope error** $|\varepsilon| \leq 3\%$, with $\varepsilon = (|\dot{M}| - \dot{M}_{\text{cmd}})/\dot{M}_{\text{cmd}}$ and $\dot{M}$ the least-squares slope over the window. A 3% error in $\dot{M}$ perturbs $C_1 = K \dot{M}$ by 3% — about $7\times$ below the $\pm 20\%$ level the sensitivity analysis shows harmless.
2. **Window samples** $N_{\text{full}} \geq 38$: a necessary condition for any onset position. Because the window opens at $|M| = 10$ mN m while $M_{\text{crit}} \gtrsim 0.4$ N m, the onset cannot occur earlier than $\hat{N}_{\text{pre}} = M_{\text{crit}}^{\text{min}}/(\dot{M} T_s) \geq 30$ samples into the window even at the fastest ramp; adding the $N_{\text{seg}} = 8$ post-onset fit minimum gives 38.
3. **Linearity RMSE** ≤ 30 mN m about the run’s own best-fit line: a fault detector for aborted/stepped ramps, set at $10\times$ the execution noise floor and above the healthy-run maximum.

The gates run *before* calibration so a poorly executed ramp cannot contaminate the dataset-coupled constraint $C_1 = K \dot{M}$.

6 Onset detection

With (C_2, K) pinned, detection reduces to an exhaustive sweep of the single remaining unknown — the onset index. Each candidate is a plain arithmetic evaluation: no optimizer, no initial guess, no seed model.

Algorithm 1 COSH onset detection (single run)

Require: samples $\{(t_i, \omega_i)\}$ on the excitation window $[i_{\text{start}}, i_{\text{end}}]$; moment $M(t)$; calibrated (C_2, K) ; measured ramp slope M ; margin $N_{\text{seg}} = 8$

Ensure: onset index j^* ; critical moment M_{crit}

- 1: $C_1 \leftarrow K\dot{M}$ ▷ onset conditions (2) fix the amplitude
 - 2: **for** $j = i_{\text{start}} + N_{\text{seg}}, \dots, i_{\text{end}} - N_{\text{seg}}$ **do** ▷ exhaustive sweep of the single unknown
 - 3: $C \leftarrow \text{median}\{\omega_i : i_{\text{start}} \leq i < j\}$ ▷ baseline by continuity
 - 4: $\text{cost}(j) \leftarrow \sum_{i=i_{\text{start}}}^{j-1} (\omega_i - C)^2 + \sum_{i=j}^{i_{\text{end}}} (\omega_i - [C_1(\cosh C_2(t_i - t_j) - 1) + C])^2$
 - 5: **end for**
 - 6: $j^* \leftarrow \arg \min_j \text{cost}(j)$; $M_{\text{crit}} \leftarrow M(t_{j^*})$
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Two implementation details. *Baseline by median.* The pre-onset gyro trace is heavy-tailed (propeller vibration, spin-up spikes), and — more importantly — when a candidate j overshoots the true onset, the pre-segment tail contains the beginning of the hyperbolic rise. A mean baseline drifts with that leaked rise and partially cancels the residual it should expose, flattening the cost landscape; the median stays at the rest level until half the segment is contaminated, so misplaced candidates are penalized sharply and the minimum stays well defined. It is also a one-shot, seedless statistic, consistent with the no-optimizer design. An ablation rerunning the full pipeline — including the (C_2, K) calibration — with the mean instead (`analysis/baseline_stat_ablation.py`) leaves the run gating unchanged and 62/132 onsets identical, but biases the remainder toward *earlier* onsets, shrinking the directional means in magnitude by up to 30 mNm on the noisiest datasets. The route is indirect. At *fixed* (C_2, K) a baseline biased toward the motion direction *delays* the detected onset: the post-onset residual $C_1[\cosh(C_2(t - t_{\text{crit}}^{\text{real}})) - \cosh(C_2(t - t_{\text{crit}}^{\text{est}}))] - C$ can only vanish with a later split, and an oppositely biased baseline advances it. The dominant effect, however, is that the leaked-rise flattening of the cost landscape moves the CV-selected calibration to a different point on the (C_2, K) ridge — e.g. $C_2 : 8.00 \rightarrow 5.38$, $K : 0.060 \rightarrow 0.150$ on the most affected dataset — whose net effect is the earlier-onset magnitude shrink; the ramp-invariance CV itself is essentially unchanged. The pivot-free offset M_{ff} moves by at most 4.2 mNm (0.13 mm), well below the run-to-run scatter — the median removes a systematic bias and tunes nothing. A second ablation pins $C = 0$, the ideal-sensor value implied by the onset conditions: the measured pre-onset baseline is in fact nonzero (gyro bias plus residual spin-up motion; $|C|$ median 3.1, p90 14.2, max 30.6 mrad/s), yet the estimate barely moves (71/132 onsets identical, M_{ff} shift ≤ 1.3 mNm). The estimated baseline is therefore not load-bearing on this dataset; it is kept because it makes the sweep invariant to the gyro bias — a vehicle- and temperature-dependent quantity that need not stay this small — at zero parameter cost. Nor does C need the bias to be constant in time: it is re-estimated per run, so only *within-window* drift (windows last 1–10 s) matters. Measured over the guaranteed pre-onset segment ($|M| < 0.3$ N m, spans ≤ 3.0 s), the baseline wanders by 3.4 mrad/s (median; p90 24, max 68) — an upper bound that conflates true sensor drift with real elastic pre-onset motion (`analysis/bias_drift.py`). This wander is present in the very data on which the $C = 0$ bound and the rate-invariance certificates were earned, and its structure is covered by the perturbation absorption: the constant part goes to C , a τ -linear trend is absorbed by the span projection, and only the residual curvature reaches ρ . *Sweep margin.* The margin $N_{\text{seg}} = 8$, applied at both window ends, guarantees every candidate split leaves at least N_{seg} samples in each segment; it excises only degenerate candidates: near $j = i_{\text{start}}$ the baseline would be estimated from a handful of samples, and near $j = i_{\text{end}}$ the post-onset arc is too short for the cosh branch to be falsifiable (at $j = i_{\text{end}}$ its cost is nearly an empty sum). Since physically feasible onsets lie at least $\hat{N}_{\text{pre}} \geq 30$ samples inside the window (Sec. 5) and the window closes only after the response is well developed, the sweep remains exhaustive over every feasible onset.

Remark 1 *Because the model curve is fully determined before the post-onset data are seen, goodness of fit is a prediction test, not flexibility: across all gate-passing runs the normalized post-onset residual is $\approx 6\%$ (gyro-noise dominated) and invariant over the twelvefold ramp-rate range — the empirical certificate that the response stays in the cosh family.*

7 Calibration of the effective constants

Algorithm 1 presupposes (C_2, K) . Although both have geometric expressions (3), they are *not* computed from geometry: z_{CoM} and especially J_P are not precisely known a priori, and — more importantly — the constants that make (3) reproduce the measured response are *effective* values that absorb the state-linear contributions of the unmodeled effects (the local slope of the gravity nonlinearity and the ground-effect spring). Both are calibrated once per dataset.

Joint residual minimization is not a usable criterion in practice. On the fitted window the amplitude and the onset are nearly interchangeable under noise: a smaller amplitude with an earlier onset and a larger amplitude with a later onset produce post-onset curves that differ most near the onset itself, where the response is quadratic, $\frac{1}{2}C_1C_2^2\tau^2$, and of the order of the gyro noise. In the noiseless limit the post-onset shape determines all three parameters uniquely — the trade-off is an ill-conditioning statement, not a non-identifiability one — but under gyro noise the summed residual is nearly flat along this amplitude–onset direction: a *shallow ridge* in the (C_2, K) plane, observed directly on the data (the objective is piecewise-constant in the integer onset index, gradient-based and simplex optimizers stall on it, and a population-based global search costs about $4\times$ more for no measurable gain). Per-run estimates — and with them the detected critical moments — scatter along the ridge. The calibration should therefore be read as *variance suppression*: choosing (C_2, K) once per dataset to minimize the dispersion of M_{crit} across ramp rates removes precisely that scatter, which is where the practical strength of the method lies.

The scatter direction is faint in the residual but loud in the deliverable: moving along the trade-off shifts each detected onset in *time*, and since $M_{\text{crit}} = M(t_{\text{on}})$ with M ramping at $\dot{M}$, an onset-time bias δt common across runs biases the critical moment by $\dot{M}\delta t$ — growing linearly with the ramp rate. A wrong pair thus *fans out* the identified M_{crit} across the twelvefold rate range, while the correct pair leaves it rate-invariant: exactly the direction a rate-consistency criterion sees best. We therefore calibrate by *physical self-consistency*: a static threshold must not depend on how fast it was approached. With runs at ramp rates $r \in \{0.10, \dots, 1.20\}$ N m/s,

$$\text{score}(C_2, K) = \sum_{\text{dir} \in \{+, -\}} \text{CV}\left(\{M_{\text{crit}}^{(r)}(C_2, K)\}_r\right), \quad \text{CV} = \text{std}/|\text{mean}|, \quad (4)$$

where $M_{\text{crit}}^{(r)}$ is the output of Algorithm 1 on the run with commanded rate r .

Search ranges. The admissible box is set so that it covers every physically plausible constant with margin. For $K = 1/(Wz_{\text{CoM}})$, the coarse range $K \in [0.10, 0.50]$ (N m)^{−1} corresponds to $z_{\text{CoM}} \in [0.06, 0.32]$ m at the vehicle weight — bracketing the plausible CoM heights — with the refinement allowed to move within the wider clip box $[0.05, 0.70]$ (z_{CoM} up to 0.63 m). For $C_2 \in [3, 8]$ rad/s, the implied pivot inertia $J_P = 1/(KC_2^2)$ spans roughly $[0.03, 1.1]$ kg m², bracketing the parallel-axis estimate $J + mz_{\text{CoM}}^2 \approx 0.2\text{--}0.35$ kg m² by a factor of several on both sides.

Algorithm 2 Ramp-invariance consistency score

Require: set $\mathcal{G}$ of gated runs (both tip directions, spanning the rate set $\{0.10, \dots, 1.20\}$ N m/s); candidate pair (C_2, K)

Ensure: consistency score s

▷ lower = M_{crit} more rate-invariant

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1: for dir ∈ {+, −} do
2:    $\mathcal{G}_{\text{dir}} \leftarrow \{\rho \in \mathcal{G} : \text{dir}(\rho) = \text{dir}\}$ 
3:   for each run  $\rho \in \mathcal{G}_{\text{dir}}$  do
4:      $M_{\text{crit}}^{(\rho)} \leftarrow$  Algorithm 1 with  $(C_2, K)$  pinned
5:   end for
6:    $\text{CV}_{\text{dir}} \leftarrow \text{std}(\{M_{\text{crit}}^{(\rho)}\}_{\rho \in \mathcal{G}_{\text{dir}}}) / |\text{mean}(\{M_{\text{crit}}^{(\rho)}\}_{\rho \in \mathcal{G}_{\text{dir}}})|$ 
7: end for
8: return  $s \leftarrow \text{CV}_+ + \text{CV}_-$ 

```

▷ Eq. (4): a static threshold must not depend on the approach rate

Note that the ramp rate never appears explicitly in the score: each direction’s gated runs span the full rate set, so the within-direction dispersion of the per-run critical moments *is* the rate dependence (plus repeatability scatter, which the score harmlessly co-penalizes). Should a rate be repeated, each run enters individually.

Algorithm 3 Effective-constant calibration (coarse-to-fine)

Require: gated runs; box $C_2 \in [3, 8]$ rad/s, $K \in [0.05, 0.70]$ (N m)⁻¹

Ensure: calibrated pair (C_2^*, K^*)

- 1: **coarse:** evaluate Algorithm 2 on the 11×11 grid $C_2 = 3.0:0.5:8.0$, $K = 0.10:0.04:0.50$
 - 2: $(C_2^b, K^b) \leftarrow \arg \min$ over the coarse grid
 - 3: **fine:** evaluate Algorithm 2 on a 9×9 uniform grid over $[C_2^b \pm 0.5] \times [K^b \pm 0.04]$, clipped to the box ▷ resolution 0.125 rad/s, 0.01 (N m)⁻¹
 - 4: $(C_2^*, K^*) \leftarrow \arg \min$ over the fine grid
-

Coarse-to-fine procedure. Algorithm 2 states the consistency criterion as a stand-alone routine — it wraps Algorithm 1 and returns the directional sum of coefficients of variation — and Algorithm 3 performs the search over it. The coarse scan evaluates (4) on an 11×11 grid ($\Delta C_2 = 0.5$ rad/s, $\Delta K = 0.04$ (N m)⁻¹; 121 evaluations). A single refinement pass then re-scans a 9×9 uniform grid spanning $\pm$ one coarse step about the coarse minimizer (± 0.5 rad/s, ± 0.04 (N m)⁻¹, clipped to the box), i.e. final resolutions of 0.125 rad/s and 0.01 (N m)⁻¹ — so the reported constants are not quantized to the coarse step. One pass suffices because the objective is flat along the (C_2, K) ridge; it is also *piecewise constant* (the onset is an integer index, so small parameter changes leave M_{crit} unchanged), which is why gradient-based or simplex optimizers stall on it, and a population-based global search (differential evolution) was found to cost about 4× more for no measurable gain. During calibration the onset sweep of Algorithm 1 is subsampled by a stride of 3 samples for speed; the final identification runs at full resolution with the selected pair. Each evaluation is an arithmetic sweep, so no iterative optimizer is involved anywhere in the pipeline.

Remark 2 *The objective (4) retains a shallow valley, so the individual constants are only loosely determined: $Wz_{\text{CoM}} = 1/K$ and $J_P = 1/(KC_2^2)$ are order-of-magnitude sanity checks, not precision measurements. What the criterion pins down — by construction — is the identified onset: M_{crit} is robust along the valley, and a $\pm 20\%$ mis-specification of either constant moves the identified CoM by less than 0.9 mm.*

8 Aggregation and deliverables

Directional means over the gated runs give $\bar{M}_+$ and $\bar{M}_-$. The take-off feed-forward moment is the *pivot-free average*

$$M_{\text{ff}} = \frac{1}{2} (\bar{M}_+ + \bar{M}_-), \quad (5)$$

in which the pivot-arm terms $\pm(W - f)l_p$ — and any thrust-proportional contamination acting through the arm l_p , ground effect included — cancel by antisymmetry, while landing-gear asymmetry is deliberately *retained*: the same contact-phase moment acts at lift-off, where M_{ff} compensates it. The CoM offset follows from the moment-to-offset conversion of M_{ff} and is used for validation against load-cell ground truth only; M_{ff} itself is the quantity applied in flight. A roll excitation (M_x) senses the y -offset via $M_{\text{ff},x} = +W y_{\text{off}}$; a pitch excitation (M_y) senses the x -offset via $M_{\text{ff},y} = -W x_{\text{off}}$.

9 Comparison with per-run estimators: NLS, CPD, CUSUM

Four per-run alternatives were run on *identical* inputs — same signals, excitation windows, allocator caps and onset-free gates (the gated run set is method-independent because the gates use the moment trace only), 132 runs in total (`analysis/nls_comparison.py`):

- **NLS** — the predecessor of Algorithm 1: per-run nonlinear least squares (SCIPY `least_squares`, trust-region reflective), free (C_1, C_2, C) with C_2 bounded to $[3, 8]$ rad/s, onset swept locally around a quadratic-model seed;
- **CPD_N / CPD_R** — optimal single change-point detection (`ruptures`; with one break the search over all admissible splits is exact) under the Gaussian mean+variance cost and the nonparametric RBF-kernel cost, respectively;
- **CUSUM** — the tabular sequential detector with allowance $k = 2\sigma$ against the pre-onset vibration.

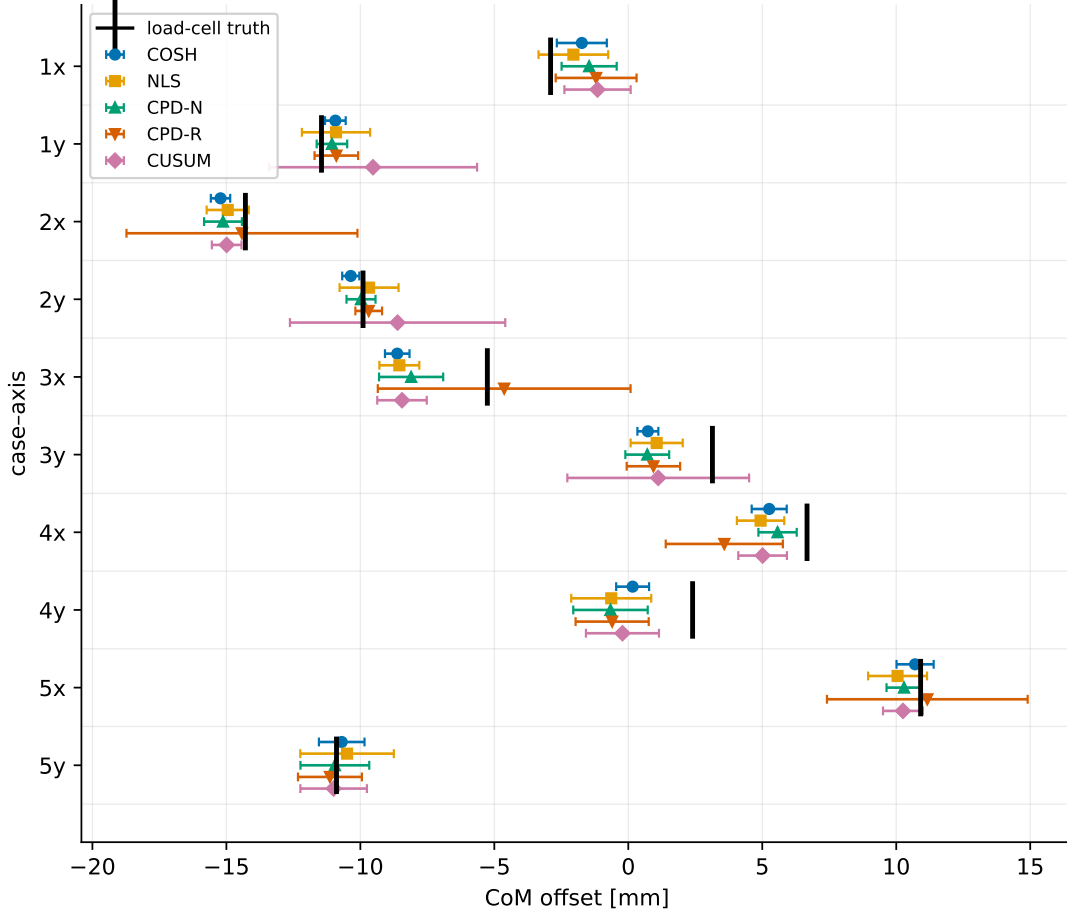


Figure 1: Estimated CoM offset with 95% confidence intervals for the five estimators, against the load-cell truth (black tick), per case-axis pair. COSH (blue) has the narrowest interval in 9 of 10 pairs (± 0.3 – 0.9 mm); the nonparametric detectors blow up to ± 4 – 5 mm on the pairs where they misplace onsets (CPD_R at 2x, 3x, 5x; CUSUM at 1y, 2y, 3y). Marker shapes duplicate the color coding.

Figure 1 is the readable summary: per case-axis pair it shows each estimator’s CoM offset with its 95% confidence interval (Welch t on the pivot-free average, run-to-run scatter) against the load-cell truth. The full data sit in the tables: Table 2 compares the deliverable — the CoM offset from the pivot-free average (5) — against the load-cell ground truth; Table 3 lists the underlying directional critical-moment means, pivot-free offsets and their confidence intervals; and Table 4 compares the ramp-rate dispersion of the directional critical moments.

Three observations. (i) *The deliverable is detector-agnostic*: against the load-cell truth all five estimators land at RMS 1.6–1.8 mm (Table 2, bottom rows) — the pivot-free average (5) cancels each method’s largely common-mode onset bias between the + and – directions, and 5–7 runs per direction average the rest. That the final CoM offset does not depend on the detection methodology is itself a validation certificate — and Fig. 2 sharpens it: the residual errors that do remain (cases 3–4) are shared by all five estimators, placing them in the vehicle-contact geometry rather than in any onset detector. (ii) *Run-level robustness is where the methods separate*: COSH has the lowest CV in 16/20 direction groups (CPD_N takes the other 4), median 2.4% against 3.9–7.1%, and its worst group is 6.5% where the alternatives reach 15–27% — the nonparametric detectors (CPD_R, CUSUM) occasionally misplace the onset badly on short fast-ramp windows. Equivalently, the 95% confidence interval on the CoM offset is ± 0.3 – 0.9 mm under COSH — the narrowest in 9 of 10 case-axis pairs — against up to ± 4 – 5 mm for the nonparametric detectors (Fig. 1). The paired per-run critical moments (NLS vs. COSH) differ by 27.6 mNm at the median but up to 217 mNm — the ridge of Sec. 7 realized run by run. (iii) *The biases carry signatures*: NLS sits 2.6% low in mean $|M|$ (the early-onset drift of the free amplitude-onset fit), while CPD_N and CUSUM sit 3.2% and 3.1% high — detection lag, since a distribution change must accumulate post-onset evidence before it is declared; both signatures largely cancel in (5). COSH needs

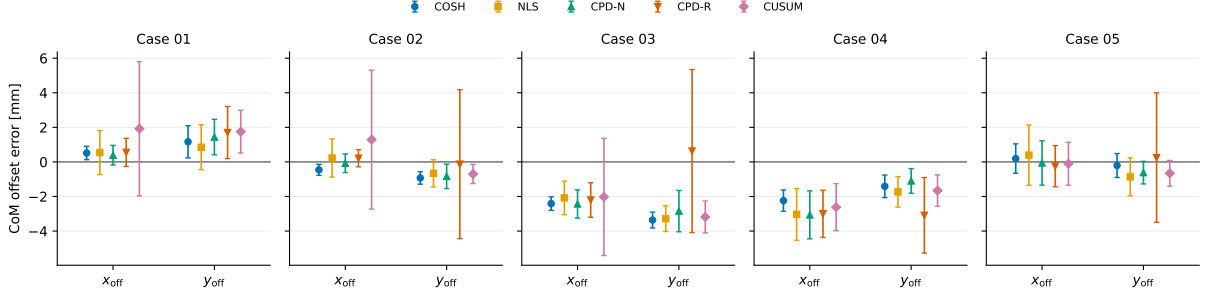


Figure 2: The same data folded about the truth: CoM offset *error* (estimate – truth) with 95% confidence intervals, grouped by component within each case (x_{off} from the pitch excitation, y_{off} from the roll excitation). The residual biases of cases 3 and 4 are *common to all five estimators* — a vehicle–contact asymmetry, not a detection artifact — while the estimator-to-estimator differences stay within about ± 1 mm.

case	axis	λ_{truth} [mm]	estimated CoM offset $\hat{\lambda} \pm \text{CI}_{95}$ [mm]				
			COSH	NLS	CPD _N	CPD _R	CUSUM
1	x	−2.90	−1.73 ± 0.93	−2.05 ± 1.30	−1.46 ± 1.03	−1.20 ± 1.51	−1.15 ± 1.24
1	y	−11.45	−10.93 ± 0.39	−10.91 ± 1.27	−11.06 ± 0.57	−10.90 ± 0.81	−9.53 ± 3.88
2	x	−14.29	−15.22 ± 0.36	−14.95 ± 0.79	−15.13 ± 0.71	−14.42 ± 4.31	−14.99 ± 0.55
2	y	−9.90	−10.36 ± 0.32	−9.67 ± 1.10	−9.98 ± 0.54	−9.69 ± 0.50	−8.61 ± 4.02
3	x	−5.26	−8.62 ± 0.46	−8.54 ± 0.74	−8.11 ± 1.20	−4.63 ± 4.72	−8.45 ± 0.92
3	y	+3.14	+0.73 ± 0.39	+1.06 ± 0.97	+0.71 ± 0.82	+0.94 ± 1.00	+1.12 ± 3.39
4	x	+6.67	+5.26 ± 0.65	+4.94 ± 0.88	+5.57 ± 0.72	+3.58 ± 2.19	+5.01 ± 0.91
4	y	+2.40	+0.16 ± 0.61	−0.64 ± 1.49	−0.66 ± 1.39	−0.60 ± 1.36	−0.22 ± 1.36
5	x	+10.91	+10.70 ± 0.69	+10.05 ± 1.10	+10.29 ± 0.65	+11.16 ± 3.75	+10.25 ± 0.74
5	y	−10.89	−10.70 ± 0.85	−10.50 ± 1.75	−10.95 ± 1.28	−11.14 ± 1.19	−11.00 ± 1.24
median err [mm]			1.05	0.86	0.97	0.59	1.70
RMS err [mm]			1.64	1.72	1.67	1.65	1.82
max err [mm]			3.36	3.28	3.06	3.09	3.19

Table 2: CoM offset from the pivot-free average (5) against the load-cell ground truth (manuscript Table 7; per-case weight 3.066 kg for case 1, 3.220 kg otherwise), for the five onset estimators on identical inputs. $M_{\text{ff}} = \pm W\hat{\lambda}$ per the axis mapping of Sec. 8.

no per-run tuning (CUSUM’s allowance k , CPD’s minimum segment) and a *single* run already sits within a few percent of its directional mean — which is what enables run-count reduction and per-run gating diagnostics.

10 Reproducibility

Every number in this document is reproducible from the repository:

- `critical_value_getter_piecewise.py` — pipeline, gates, caps, calibration, Algorithm 1;
- `analysis/ramp_quality.py` — gates;
- `analysis/sensitivity.py` — $\pm 20\%$ robustness;
- `analysis/cosh_fidelity.py` — prediction-test residuals;
- `analysis/excitation_angle_design.py` — rate/cutoff design;
- `analysis/ge_linearity.py`, `analysis/ge_trajectory.py` — ground-effect bracket;
- `analysis/tiltcap_ablation.py` — window ablation;
- `analysis/baseline_stat_ablation.py` — median/mean/zero baseline;
- `analysis/bias_drift.py` — in-window baseline drift;

method		1x	1y	2x	2y	3x	3y	4x	4y	5x	5y
COSH	$\bar{M}_-$	-1.054	-0.422	-1.829	-0.666	-1.542	-0.983	-1.136	-0.998	-1.019	-0.635
	$\bar{M}_+$	+0.950	+1.079	+0.867	+1.320	+0.997	+0.937	+1.468	+0.988	+1.695	+1.310
	M_{ff}	-0.052	+0.329	-0.481	+0.327	-0.272	-0.023	+0.166	-0.005	+0.338	+0.338
	CI_{95}	± 0.028	± 0.012	± 0.011	± 0.010	± 0.014	± 0.012	± 0.021	± 0.019	± 0.022	± 0.027
NLS	$\bar{M}_-$	-1.018	-0.399	-1.792	-0.637	-1.573	-0.990	-1.151	-0.928	-0.989	-0.577
	$\bar{M}_+$	+0.894	+1.056	+0.848	+1.248	+1.033	+0.923	+1.463	+0.969	+1.624	+1.241
	M_{ff}	-0.062	+0.328	-0.472	+0.306	-0.270	-0.033	+0.156	+0.020	+0.317	+0.332
	CI_{95}	± 0.039	± 0.038	± 0.025	± 0.035	± 0.024	± 0.031	± 0.028	± 0.047	± 0.035	± 0.055
CPD _N	$\bar{M}_-$	-1.048	-0.464	-1.852	-0.703	-1.573	-1.059	-1.164	-1.005	-1.033	-0.640
	$\bar{M}_+$	+0.960	+1.130	+0.896	+1.333	+1.061	+1.014	+1.516	+1.047	+1.683	+1.332
	M_{ff}	-0.044	+0.333	-0.478	+0.315	-0.256	-0.022	+0.176	+0.021	+0.325	+0.346
	CI_{95}	± 0.031	± 0.017	± 0.022	± 0.017	± 0.038	± 0.026	± 0.023	± 0.044	± 0.021	± 0.041
CPD _R	$\bar{M}_-$	-1.023	-0.449	-1.680	-0.702	-1.339	-1.055	-1.179	-0.986	-0.932	-0.625
	$\bar{M}_+$	+0.950	+1.105	+0.769	+1.314	+1.046	+0.995	+1.406	+1.024	+1.637	+1.328
	M_{ff}	-0.036	+0.328	-0.455	+0.306	-0.146	-0.030	+0.113	+0.019	+0.353	+0.352
	CI_{95}	± 0.045	± 0.024	± 0.136	± 0.016	± 0.149	± 0.032	± 0.069	± 0.043	± 0.118	± 0.038
CUSUM	$\bar{M}_-$	-1.046	-0.480	-1.861	-0.710	-1.608	-1.029	-1.181	-1.044	-1.048	-0.656
	$\bar{M}_+$	+0.977	+1.054	+0.914	+1.254	+1.075	+0.958	+1.498	+1.058	+1.695	+1.351
	M_{ff}	-0.035	+0.287	-0.473	+0.272	-0.267	-0.035	+0.158	+0.007	+0.324	+0.347
	CI_{95}	± 0.037	± 0.117	± 0.018	± 0.127	± 0.029	± 0.107	± 0.029	± 0.043	± 0.024	± 0.039

Table 3: Directional critical-moment means $\bar{M}_{\pm}$, pivot-free offset M_{ff} and its 95% confidence interval (Welch t) [Nm] per estimator. Columns are case-axis pairs (e.g. 1x = case 1, roll). Note the signed signatures relative to COSH: NLS magnitudes sit low (early onsets), CPD_N/CUSUM high (detection lag), while M_{ff} stays nearly method-invariant.

- `analysis/nls_comparison.py` — Tables 2-4.

case	axis	COSH		NLS		CPD _N		CPD _R		CUSUM	
		CV ₋	CV ₊	CV ₋	CV ₊	CV ₋	CV ₊	CV ₋	CV ₊	CV ₋	CV ₊
1	x	3.8	5.7	6.4	7.7	4.4	6.1	7.3	8.6	5.9	6.7
1	y	2.4	2.2	8.8	7.2	1.8	3.3	5.2	4.4	5.2	24.0
2	x	1.2	1.5	2.5	4.2	2.5	2.4	15.5	23.6	1.8	2.6
2	y	1.2	1.6	4.9	5.8	2.8	2.5	3.1	2.0	3.1	21.9
3	x	1.7	2.4	2.4	4.1	4.9	4.2	23.9	8.5	3.1	4.7
3	y	1.7	2.6	3.2	7.0	4.4	4.1	4.4	6.0	12.4	22.8
4	x	3.4	2.2	3.8	3.5	4.0	1.6	5.6	10.3	3.9	3.5
4	y	2.9	3.7	10.3	6.1	7.5	7.2	7.8	6.9	6.7	7.3
5	x	3.9	2.1	7.2	2.6	3.6	2.0	27.4	3.0	4.0	2.3
5	y	6.5	2.9	15.2	5.4	10.1	3.8	9.5	3.8	9.3	4.0
median / worst [%]		2.4 / 6.5		5.6 / 15.2		3.9 / 10.1		7.1 / 27.4		4.9 / 24.0	

Table 4: Ramp-rate dispersion (CV [%]) of the directional critical moments under the five estimators. Bold marks the lowest CV in each of the 20 direction groups: COSH in 16, CPD_N in 4, the rest in none. The underlying directional means are in Table 3.